Vision–Language Learning for Remote Sensing: From Change Detection to Super-Resolution

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***Abstract*—This summary presents three advances in multimodal learning for remote sensing. All integrate vision–language models such as CLIP to bridge the semantic gap in traditional remote sensing analysis. Papers of MdaCD, Text-RSSR and TG-RST are avaliable.**

***Index Terms*—Remote sensing, remote sensing change detection, remote sensing super-resolution, multimodal learning**

I. Multimodal Difference Augmentation for Change Detection

FIGURE 1 illustrates the core motivation of MdaCD: in remote sensing change detection, textual descriptions generated by CLIP often emphasize unchanged regions while overlooking subtle but meaningful changes. This occurs because unchanged areas dominate the image and domain shifts distort textual alignment. As shown, CLIP's descriptions misinterpret scene variations, highlighting harbors instead of industrial growth. MdaCD addresses this by guiding CLIP's attention toward changed regions through CLIP-Guided Masking, enabling more accurate multimodal alignment and differential feature learning. *This work is published on IEEE Transactions on Geoscience and Remote Sensing.*

II. Text-Conditioned Remote Sensing Super-Resolution

Figure 2 highlights the motivation of Text-RSSR: during satellite imaging, transmitting full high-resolution data to Earth is costly, while low-resolution images lose essential details. Text descriptions, however, require negligible storage and can preserve semantic and structural information of the original high-resolution scene before downsampling. By integrating these captions as auxiliary inputs, Text-RSSR leverages textual semantics to guide visual reconstruction, enabling the recovery of fine spatial details and improving the fidelity of remote sensing super-resolution. *This work is anonymously submitted to a computer vision conference.*

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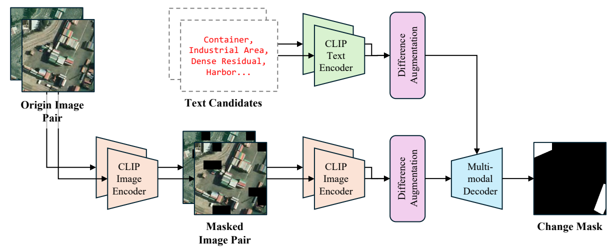

Fig. 1: Overall architecture of MdaCD.

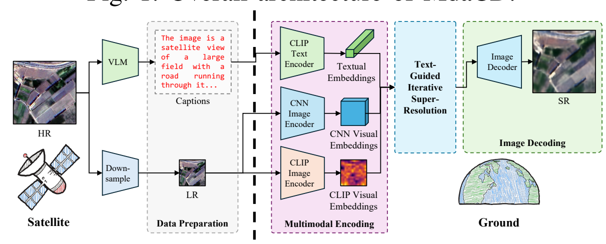

Fig. 2: Overall architecture of Text-RSSR.

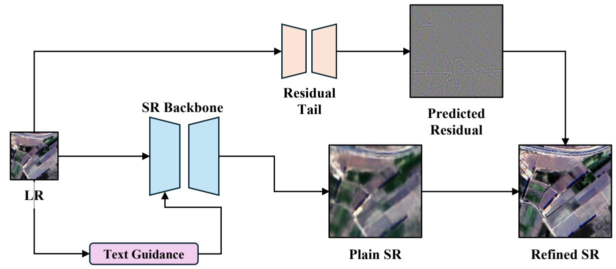

Fig. 3: Overall architecture of TG-RST.

III. Remote Sensing Super-Resolution With Tail Fine-Tuning

Figure 3 shows the motivation of TG-RST: existing remote sensing super-resolution networks often leave behind consistent, structured residual errors, because the low-resolution input and the remote sensing domain introduce strong downsampling and low- and high-resolution domain shifts that are difficult to fully model. TG-RST addresses this limitation by coupling a tailless text-guided RSSR backbone with a lightweight multi-stage Tail refinement strategy. In this way, TG-RST uses text to improve semantic alignment and structural consistency while using residual learning to explicitly restore the missing high-frequency components. *This work is submitted to IEEE Transactions on Geoscience and Remote Sensing.*